

Qatar-1b

R-2720

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110830 · created 2026-09-01T02:05:15+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455803.83507 +/- 0.0004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.044
Transit depth (Rp/Rs)^2	3.55 +/- 1.67 [%]
Orbital Inclination (inc)	89.03 +/- 2.4
Airmass coefficient 1 (a1)	1.202 +/- 0.047
Airmass coefficient 2 (a2)	-0.112 +/- 0.053
Scatter in the residuals of the lightcurve fit is	3.97 %
Best Comparison Star	#3 - [486, 377]
Optimal Aperture	3.31
Optimal Annulus	6.05
Transit Duration (day)	0.0838 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.188 $\pm$ 0.044 vs 0.1463 (deviation 0.95 σ)
Duration fit vs published	0.0838 d vs 0.0692 d (deviation 2.66 σ)
Mid-time O-C	-0.9 min
Depth SNR	4.3
Residual scatter	3.97 %

Light curve

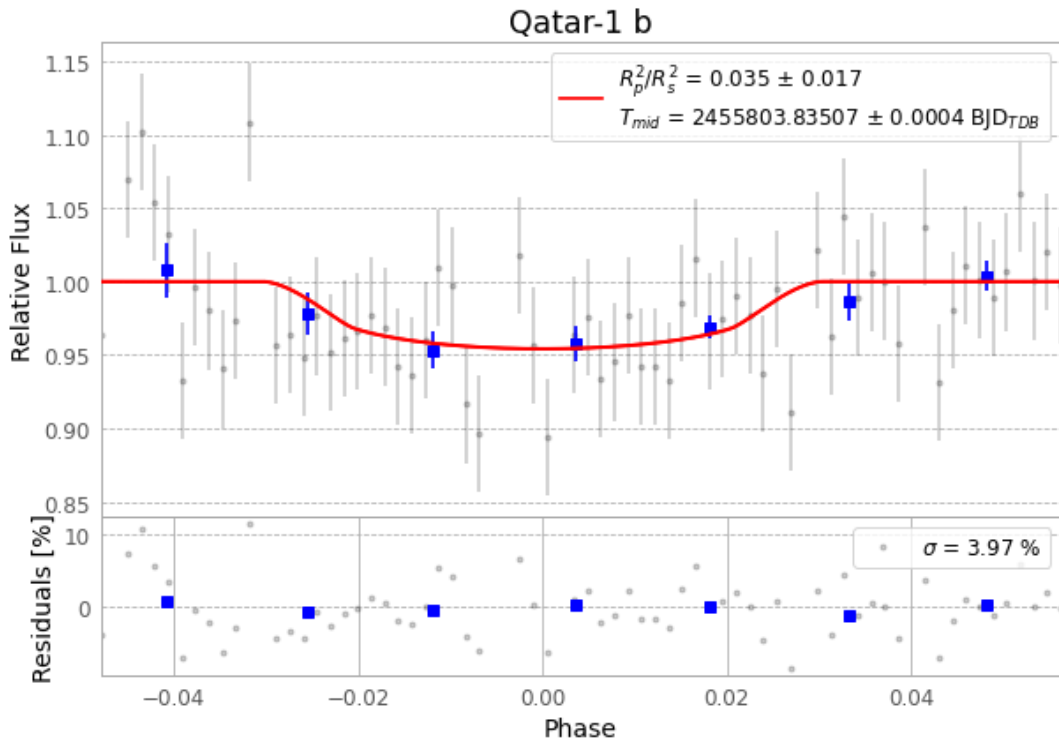


Figure 1 — FinalLightCurve_Qatar-1 b_2011-08-29.png (SHA-256 56b070d26eef58ab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 300], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9b5c8b5a7392081d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1110830062112.FITS → Qatar-1110830095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110830.log (07b83a570366...), FinalLightCurve_Qatar-1 b_2011-08-29.png (56b070d26eef...), FinalLightCurve_Qatar-1 b_2011-08-29.csv (23d81a8a330a...)

Compute resources

Wall time 184.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 02:05Z.